♥ JONI HENDRICKSON ♥

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Seattle-based embedded firmware engineer with experience in agricultural robotics, satellites, and IoT.

C, Embedded Rust, FreeRTOS, Zephyr	C, Rust, Scripting (Bash, Python)	SWD/JTAG, Logic Analyzer/Oscilloscope
CAN, UART, SPI, I2C, Network	Yocto, Ansible, Terraform, Tailscale	Sensors, Motors, Batteries, Cameras
OTA, Security, QEMU, HITL CI/CD	Docker, ROS2, systemd, NVIDIA Jetson	PCB bring-up, Manufacture, Debug

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Bonsai Robotics – Senior Embedded Platform Engineer ***Mar 2026 – Current***

Autonomous electric tractors and other agricultural equipment

- Developed Embedded Linux platform for robotic tractors on NVIDIA Jetson Orin using ROS2, Yocto, and Docker
- Built OpenTofu/Terraform fleet management system to configure and update network-constrained devices with Qbee
- Implemented patches for perception modules to support multiple NVIDIA library versions (TensorRT/CUDA/cuDNN)
- Organized firmware projects, including STM32 J939 CAN bus devices, GMSL cameras, LVGL display, and GPS
- Collaborated with electrical, mechanical, and manufacturing engineers to integrate complex machinery

Amazon – Embedded Software Engineer ***Sept 2021 – Dec 2025***

Amazon Go / Just Walk Out – MCU Firmware *Oct 2024 – Dec 2025*

- Developed firmware for STM32-based weight sensors, edge networks, cameras, and IoT on Linux and FreeRTOS
- Rebuilt HITL CI/CD pipeline to support rapid deployment and new network topologies
- Wrote manufacturing/test harness for a new motorized gate to evaluate PID tuning for steppers and BLDCs
- Traveled on-site to deploy and validate embedded systems, developing solutions to recurring problems

Project Kuiper – Satellite Embedded Software *Aug 2023 – Oct 2024*

- Updated and maintained a framework for modular satellite applications in Rust and C++ on Embedded Linux
- Improved IPC, telemetry, storage, and logging systems, porting key components from C++ to Rust
- Enhanced fault tolerance mechanisms, including health monitoring, watchdogs, and time synchronization
- Provided on-call support during prototype satellite launches and operations

FreeRTOS Maintainer Sept 2021 – Aug 2023

- Developed MISRA-compliant embedded C libraries for FreeRTOS, including OTA updates, HTTP, and cryptography
- Wrote firmware for AWS IoT ExpressLink, implementing OTA, MQTT, and BLE GAP/GATT over UART commands
- Built a HITL CI/CD pipeline for Linux/RPi testbeds, automating Pytest-based MCU integration tests
- Worked with ESP32, Realtek, u-blox, and other vendors for reference integrations of FreeRTOS

Avanade – Software Developer Intern ***June 2021 – Aug 2021***

- Developed interactive AltspaceVR features in Node.js, including virtual games and in-app item systems

PROJECTS

joni-on-micro.site - Microcontroller-Based Web Server ***Jan 2025***

- Built an embedded web server on an ESP32-S3 using LittleFS and esp-httpd to serve a static blog site
- Teardown 2025 hardware conference talk

cedh-decklist-database.com – Web Developer ***June 2018***

- Built a website with an AWS-powered backend for browsing and submitting Magic: The Gathering decklists

EDUCATION

University of Washington Bothell – BS Computer Science & Software Engineering ***Sept 2018 – Aug 2021***

- GPA: 3.77 | Courses: Operating Systems, Hardware and Computer Organization, Data Structures and Algorithms
- As a Teaching Assistant for CSE 180: taught data manipulation, visualization, and statistics